If structured binding

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Show to_chars without this proposal

Symantec

```
std::array<char, 10> str;

if (auto res = std::to_chars(str.data(), str.data() + str.size(),
                             format_args...))
    std::println("{}", std::string_view(str.data(), res.ptr));
else
    std::println("{}", std::make_error_code(res.ec).message());
```

Show to_chars with this proposal

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```
std::array<char, 10> str;

if (auto [ptr, ec] = std::to_chars(str.data(), str.data() + str.size(),
                                   format_args...))
    std::println("{}", std::string_view(str.data(), ptr));
else
    std::println("{}", std::make_error_code(ec).message());
```

Examples

Example 1

```
if (auto [first, last] = parse(begin(), end()); first != last)
{
    // interpret [first, last) into a value
}
```

Does this mean
last != end()
?

Why don't I write
first != last &&
std::string_view(first, last)
!= KW_ANY
?

```
struct parse_window
{
    char const *first, *last;
};




parse_window parse(char const*, char const*);
```

Adding operator bool

```
struct parse_window
{
    char const *first, *last;
    explicit operator bool() const noexcept
    {
        return first != last;
    }
};

parse_window parse(char const*, char const*);
```

Example 1

```
if (auto [first, last] = parse(begin(), end()); first != last)
{
    // interpret [first, last) into a value
}
```

Example 1 simplified with this proposal

```
if (auto [first, last] = parse(begin(), end()))
{
    // interpret [first, last) into a value
}
```

Example 1

```
if (auto [first, last] = parse(begin(), end()); first != last)
{
```

Information about the condition is spread across the components

How do I reconstruct the condition?

Example 1 simplified with this proposal

```
if (auto [first, last] = parse(begin(), end()))
{
```

Users channeled the intention with ‘if’

Example 2...?

```
if (auto result = std::to_chars(p, last, 42))
{
    // okay to proceed
}
```

Why don’t you...?

```
if (auto result = std::to_chars(p, last, 42))
{
    auto [ptr, ec] = result;
    // okay to proceed
}
```

Example 2

```
if (auto result = std::to_chars(p, last, 42))
{
    // okay to proceed
}
else
{
    // handle errors
}
```

Example 2

```cpp
if (auto result = std::to_chars(p, last, 42))
{
    auto [ptr, _] = result;
    // okay to proceed
}
else
{
    auto [ptr, ec] = result;
    // handle errors
}
```

Adding structured binding declaration in if...else body is infeasible

Why don’t you...?

```
if (auto [ptr, ec] = std::to_chars(p, last, 42); ec != std::errc{})
{
    // okay to proceed
}
else
{
    // handle errors
}
```

Why don’t you…?

```
if (auto [ptr, ec] = std::to_chars(p, last, 42); ec != std::errc{})
{
```

A single component contains information about the condition

Still want to channel the information via the complete object

Between this

```
if (auto [ptr, ec] = std::to_chars(p, last, 42); ec != std::errc{})
{
    // okay to proceed
}
else
{
    // handle errors
}
```

...and this

```
if (auto result = std::to_chars(p, last, 42))
{
    auto [ptr, _] = result;
    // okay to proceed
}
else
{
    auto [ptr, ec] = result;
    // handle errors
}
```

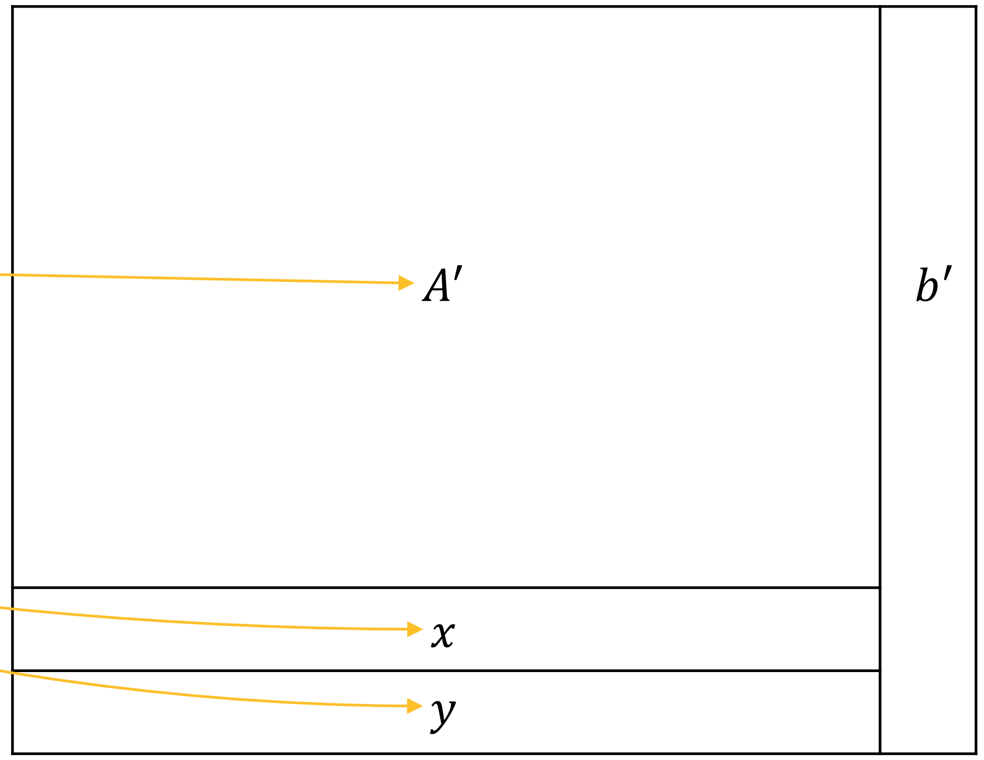
I choose

```
if (auto [ptr, ec] = std::to_chars(p, last, 42))
{
    // okay to proceed
}
else
{
    // handle errors
}
```

Example 3

iterative solver

```
struct tableau
{
    matrix Ap;
    vector bp, x, y;
};
```



Example 3

iterative solver

```
struct tableau
{
    matrix Ap;
    vector bp, x, y;
};
```

```
class Solver
{
  public:
    tableau solve();
    bool is_optimal(vector const&);
};
```

Example 3

```
for (;;)
{
    // ...
    auto [Ap, bp, x, y] = solve();

    // stop iteration
}
```

Example 3

```
for (;;)
{
    // ...
    auto [Ap, bp, x, y] = solve();

    if (is_optimal(x))
        break;
}
```

But solve() knew whether the last step reached optimal...

Example 3 improved with the proposal

```
for (;;)
{
    // ...
    if (auto [Ap, bp, x, y] = solve())
        break;
}
```

Saves the cost of reconstructing the information from the components

But solve() knew whether the last step reached optimal...

Example 4

```
if (auto [all, city, state, zip] =
        ctre::match<"(\\w+), (\\w+) (\\d+)">(s); all)
{
    return location{ city, state, zip };
}
```

Example 4

```
if (auto [all, city, state, zip] =
            ctre::match<"(\\w+), (\\w+) (\\d+)">(s); all)
{
```

All components but one have similar roles

Example 4 designed differently with the proposal

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```
if (auto [city, state, zip] = ctre2::match<"(\\w+), (\\w+) (\\d+)">(s))
{
    return location{ city, state, zip };
}
```

Technical Details

Where can I use this?

- `if constexpr`$_{opt}$ `(` *`init-statement`*$_{opt}$ *`condition`* `)` *`statement`* `else` *`statement`*
- `switch (` *`init-statement`*$_{opt}$ *`condition`* `)` *`statement`*
- `while (` *`condition`* `)` *`statement`*
- `for (` *`init-statement`* *`condition`*$_{opt}$ `;` *`expression`*$_{opt}$ `)` *`statement`*

Can decomposition be conditional?

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```
auto consume_int() -> std::optional<int>;


    if (auto [i] = consume_int()) // let e be the underlying object
    {
        // i = *e
    }
    else
    {
        // *e is not evaluated
    }
```

Can decomposition be conditional?

- No

To destructure	Abusing condition
`optional<T>`	`[x]`

Can decomposition be conditional?

Symantec

- No

To destructure	Abusing condition
`optional<T>`	`[x]`
`optional<tuple<T, U>>`	`[x, y]` ✖

Can decomposition be conditional?

- No

To destructure	Abusing condition	Pattern matching
optional<T>	[x]	let ?x
optional<tuple<T, U>>	[x, y]	let ?[x, y]

Pattern matching's solution

```
if (consume_int() match let ?i)
{
    // use(i)
}
else
{
    // has no value
}
```

When does the Boolean test happen?

- After initializing the bindings
- So that

```
if (auto [a, b, c] = fn())
{
    statements;
}
```

is equivalent to

```
if (auto [a, b, c] = fn(); underlying-object)
{
    statements;
}
```

Can I decompose an array in a condition?

- No.

Implementation Experience

Implemented in Clang

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C++

x86-64 clang (trunk)   -Wall -std=c++17

```cpp

struct format_status
{
    format_errc ec;
    char *bp;

    explicit operator bool() const noexcept
    {
        return ec == format_errc::no_error;
    }
};

format_status

int main()
{
    if (auto [ok, ptr] = readint())
    {
        printf("stopped at %p\n", ptr);
    }
}
```

warning: ISO C++17 does not permit structured binding declaration in a condition [-Wbinding-in-condition] x86-64 clang (trunk) #1

View Problem (Alt+F8)   No quick fixes available

```asm
main: # @main
  push rax
  call readint()@PLT
  test eax, eax
  jne .LBB0_2
  lea rdi, [rip + .L.str]
  mov rsi, rdx
  xor eax, eax
  call printf@PLT
.LBB0_2:
  xor eax, eax
  pop rcx
  ret
.L.str:
  .asciz "stopped at %p\n"
```

Thank you